Lecturer

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Course description and assessment

Aim This course provides a short introduction to Probability Theory and Statistics. Probability Theory is a branch of mathematics that deals with *chance* and *uncertainty*. It lays the foundation for the science of statistical inference through experimentation and data analysis - an area of crucial importance in an increasingly quantitative world.

Assessment Attendance (10%), tutorials (30%) and exam (60%).

- Attendance: Don't be late.
- Tutorials: There are 15 tutorials.
- Exam: To be announced.

An exam retake will be available with either a Fail or a Pass grade.

Textbook Some recommended books for this course are

+ *Probability and statistics for engineers and scientists* by Anthony Hayter (4th edition).
+ *Introduction to probability* by David F. Anderson.

Academic honesty, cheating and plagiarism Cheating is viewed as a serious academic offence. The University will not tolerate cheating or assisting others to cheat. Instances of cheating or plagiarism will result in a fail for this course.

Look through the Syllabus for more information.

Introduction to Probability

Tan Do

Vietnamese-German University

Lecture 1

In this lecture

- Probabilities: experiments, outcomes, sample spaces, Venn diagrams, probability values

Introduction

- Probability is a branch of mathematics that deals with chance and uncertainty.
- Probability forms the basis for statistical inference through experiment and data analysis.

Sample spaces

An experiment is a process or a procedure which produces some outcomes (possibly more than one outcome).

The set of all outcomes is called the sample space or the state space and is denoted by $\mathcal{S}$.

Example (Machine breakdowns) A machine may break down due to an electrical failure, a mechanical failure or operator misuse. When the machine is running, the engineer is uncertain what will be the cause of the next breakdown. The problem can be considered as an experiment with the sample space

$$\mathcal{S} = \{\text{electrical, mechanical, misuse}\}.$$

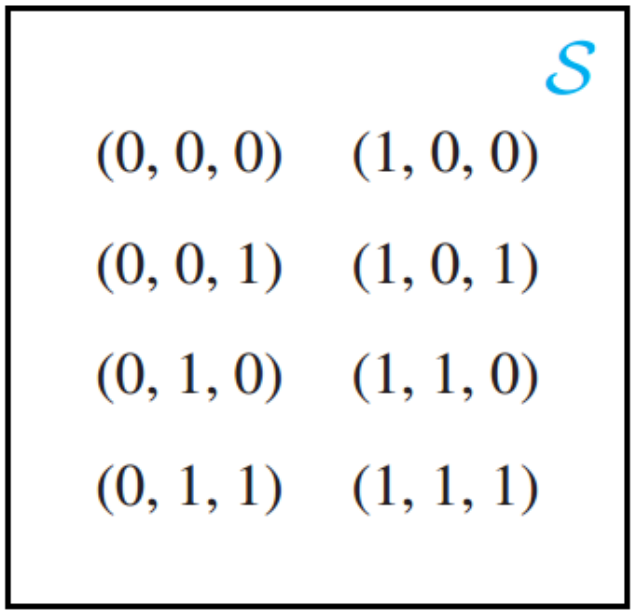
Example (Defective computer chips) A company sells computer chips in boxes of 500. Each chip can be classified as either satisfactory or defective. The number of defective chips in a particular box is uncertain, and the sample space is

$$\mathcal{S} = \{0 \text{ defectives}, 1 \text{ defective}, 2 \text{ defectives}, \ldots, 500 \text{ defectives}\}.$$

Example A manager supervises the operation of 3 power plants X, Y and Z. At any given time, each of the three plants can be classified as either generating electricity (1) or being idle (0).

The sample space for the status of the 3 plants at a particular point in time is

$$\mathcal{S} = \{(0,0,0),(1,0,0),(0,1,0),(0,0,1),(1,1,0),(1,0,1),(0,1,1),(1,1,1)\}.$$



Venn diagram

Games of chance

Games of chance commonly involve the toss of a coin, the roll of a die or the use of a pack of cards.

Coin tossing

The toss of a coin has a sample space

$$\mathcal{S} = \{\text{head, tail}\}.$$

The toss of two coins (or equivalently, the toss of a coin twice) has a sample space

$$\mathcal{S} = \{(\text{head, head}), (\text{head, tail}), (\text{tail, head}), (\text{tail, tail})\}.$$

Note that (head, tail) and (tail, head) represent 2 different outcomes.

Die rolling

A (six-sided) die has a sample space

$$\mathcal{S} = \{1, 2, 3, 4, 5, 6\}.$$

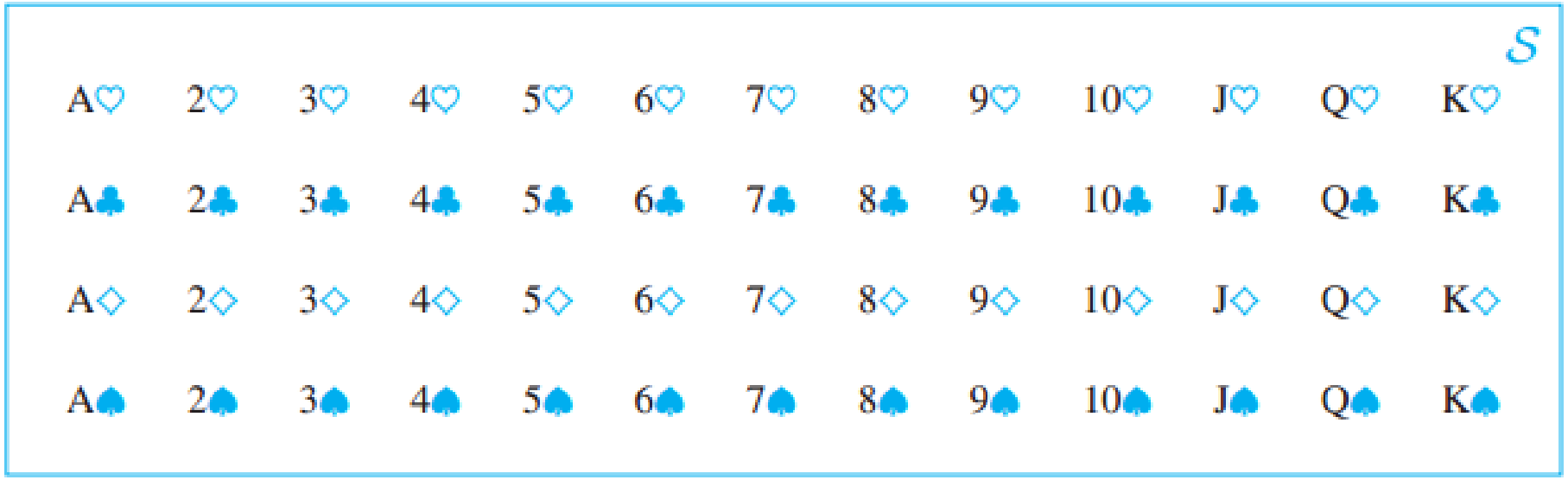
If 2 dice are rolled (or equivalently, if a die is rolled twice), then the sample space is

$\mathcal{S}$

(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Sample space for rolling 2 dice

Card playing If a card is chosen from an ordinary pack of 52 playing cards, the sample space is



Sample space for choosing 1 card

If 2 cards are drawn, then it is necessary to consider whether they are drawn *with or without replacement*.

Drawing with replacement The initial card drawn is returned to the pack and the second drawing is from a full pack of 52 cards. Altogether there will be $52 \times 52 = 2704$ elements of the sample space.

In this case, an outcome such as $(A\heartsuit, A\heartsuit)$ is possible.

$\mathcal{S}$

(A♡, A♡)	(A♡, 2♡)	(A♡, 3♡)	⋯	(A♡, Q♠)	(A♡, K♠)
(2♡, A♡)	(2♡, 2♡)	(2♡, 3♡)	⋯	(2♡, Q♠)	(2♡, K♠)
(3♡, A♡)	(3♡, 2♡)	(3♡, 3♡)	⋯	(3♡, Q♠)	(3♡, K♠)
⋮	⋮	⋮		⋮	⋮
(Q♠, A♡)	(Q♠, 2♡)	(Q♠, 3♡)	⋯	(Q♠, Q♠)	(Q♠, K♠)
(K♠, A♡)	(K♠, 2♡)	(K♠, 3♡)	⋯	(K♠, Q♠)	(K♠, K♠)

Sample space for choosing 2 cards with replacement

Drawing without replacement The initial card drawn is NOT returned to the pack and the second card is drawn from a pack of 51 cards, then the sample space will be a subset of that above.

In this case, an outcome such as $(A\heartsuit, A\heartsuit)$ is not possible.

The total number of outcomes is $2704 - 52 = 2652$.

$\mathcal{S}$

	(A♡, 2♡)	(A♡, 3♡)	⋯	(A♡, Q♠)	(A♡, K♠)
(2♡, A♡)		(2♡, 3♡)	⋯	(2♡, Q♠)	(2♡, K♠)
(3♡, A♡)	(3♡, 2♡)		⋯	(3♡, Q♠)	(3♡, K♠)
⋮	⋮	⋮		⋮	⋮
(Q♠, A♡)	(Q♠, 2♡)	(Q♠, 3♡)	⋯		(Q♠, K♠)
(K♠, A♡)	(K♠, 2♡)	(K♠, 3♡)	⋯	(K♠, Q♠)	

Sample space for choosing 2 cards without replacement

Probability values

We assign probability values to the elements of the sample space to represent how likely an outcome will occur.

Definition A set of probability values for an experiment with a sample space $\mathcal{S} = \{O_1, O_2, \ldots, O_n\}$ consists of some probabilities $p_1, p_2, \ldots, p_n$ which satisfy

$$0 \leq p_1, p_2, \ldots, p_n \leq 1$$

and

$$p_1 + p_2 + \ldots + p_n = 1.$$

We say that an outcome O_i occurs with probability p_i and write $P(O_i) = p_i$.

Example (Machine breakdowns) Suppose that the machine breakdowns occur with probability values of $P(\text{electrical}) = 0.2$, $P(\text{mechanical}) = 0.5$ and $P(\text{misuse}) = 0.3$.
Note that this is a valid probability assignment.

The probability values indicate that mechanical failures are most likely and misuse failures are more likely than electrical failures.

These values also say that in the *long run* roughly 50% of the breakdowns will be for mechanical reasons, about 20% of the breakdowns will be for electrical reasons and about 30% of the breakdowns will be attributed to operator misuse.

Question How does one know the probability assignments?
For example, in the above example, how would the engineer know that there is a probability of 0.2 that a breakdown will be due to an electrical fault and so on?

Answer In practice these probabilities would have to be estimated from a collection of data and prior experiences.
(Chapters 7 and 10 of the textbook show how to employ statistical analysis techniques to estimate probabilities of these kinds.)

Games of chance 2

Coin tossing

The probabilities for a coin toss will in general be given by

$$P(\text{head}) = p, \quad P(\text{tail}) = 1 - p$$

for some $p \in [0, 1]$.

A coin is fair if $p = 0.5$, i.e.,

$$P(\text{head}) = 0.5, \quad P(\text{tail}) = 0.5.$$

Otherwise, the coin is biased. For example, when $p = 0.4$,

$$P(\text{head}) = 0.4, \quad P(\text{tail}) = 0.6.$$

Die rolling

A fair die will have equally likely 6 outcomes. So the outcomes have equal probabilities:

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}.$$

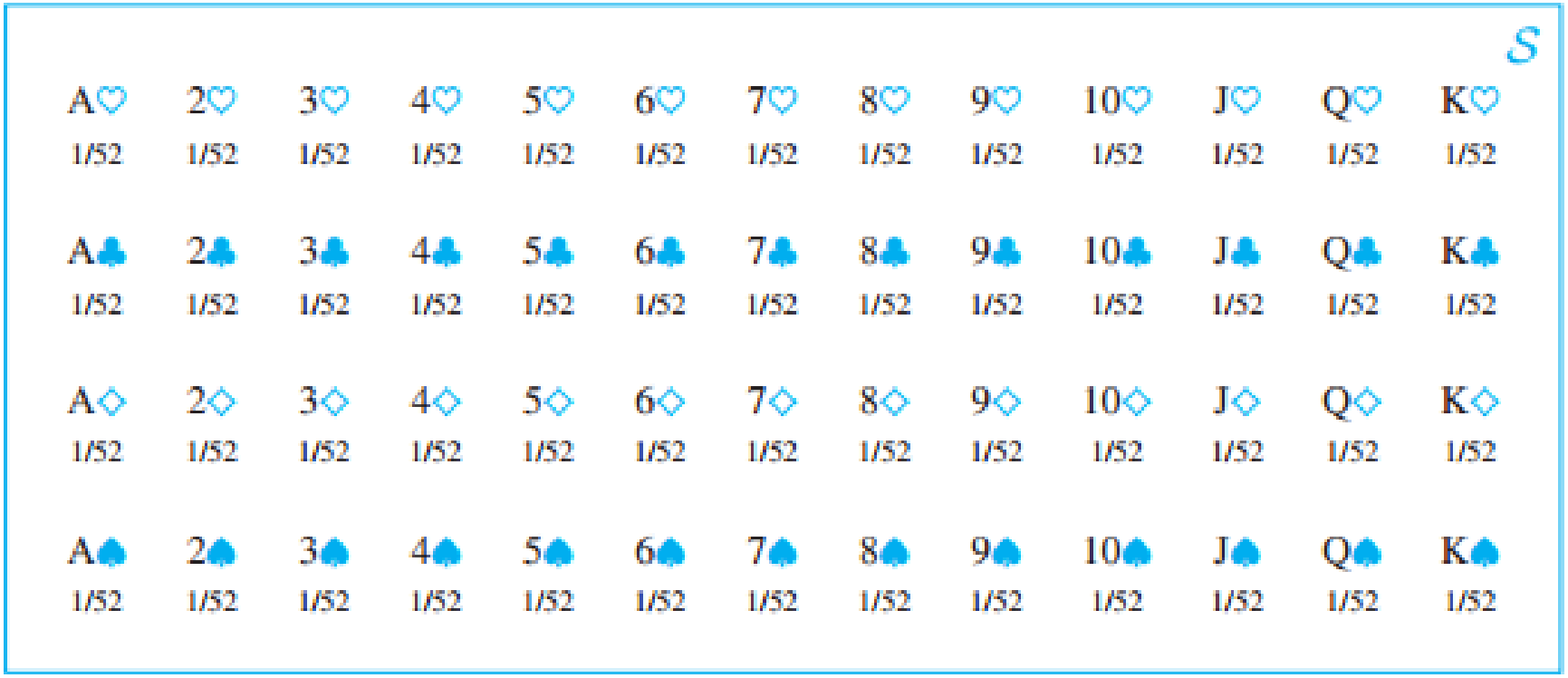
If 2 fair dice are rolled, the probability assignment is

$\mathcal{S}$

(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
1/36	1/36	1/36	1/36	1/36	1/36
(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
1/36	1/36	1/36	1/36	1/36	1/36
(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
1/36	1/36	1/36	1/36	1/36	1/36
(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
1/36	1/36	1/36	1/36	1/36	1/36
(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
1/36	1/36	1/36	1/36	1/36	1/36
(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
1/36	1/36	1/36	1/36	1/36	1/36

Probability values for rolling 2 dice

Card playing If a card is drawn at random from a pack of cards, then there are 52 possible outcomes which are equally likely. So each would be assigned a probability value of $\frac{1}{52}$.



Probability values for choosing 1 card

Drawing with replacement If both cards are drawn at random (by suitable shuffling of the pack before and between the drawings), then the 2704 elements in $\mathcal{S}$ are equally likely. So each will be assigned a probability value of $1/2074$.

$\mathcal{S}$

(A♡, A♡) 1/2704	(A♡, 2♡) 1/2704	(A♡, 3♡) 1/2704	··· ···	(A♡, Q♠) 1/2704	(A♡, K♠) 1/2704
(2♡, A♡) 1/2704	(2♡, 2♡) 1/2704	(2♡, 3♡) 1/2704	··· ···	(2♡, Q♠) 1/2704	(2♡, K♠) 1/2704
(3♡, A♡) 1/2704	(3♡, 2♡) 1/2704	(3♡, 3♡) 1/2704	··· ···	(3♡, Q♠) 1/2704	(3♡, K♠) 1/2704
⋮	⋮	⋮		⋮	⋮
(Q♠, A♡) 1/2704	(Q♠, 2♡) 1/2704	(Q♠, 3♡) 1/2704	··· ···	(Q♠, Q♠) 1/2704	(Q♠, K♠) 1/2704
(K♠, A♡) 1/2704	(K♠, 2♡) 1/2704	(K♠, 3♡) 1/2704	··· ···	(K♠, Q♠) 1/2704	(K♠, K♠) 1/2704

Probability values for choosing 2 cards with replacement

If the drawing is performed without replacement but again at random, then the 2652 elements in $\mathcal{S}$ are equally likely. So each will be assigned a probability value of $1/2652$.

$\mathcal{S}$

	(A♡, 2♡) 1/2652	(A♡, 3♡) 1/2652	··· ···	(A♡, Q♠) 1/2652	(A♡, K♠) 1/2652
(2♡, A♡) 1/2652		(2♡, 3♡) 1/2652	··· ···	(2♡, Q♠) 1/2652	(2♡, K♠) 1/2652
(3♡, A♡) 1/2652	(3♡, 2♡) 1/2652		··· ···	(3♡, Q♠) 1/2652	(3♡, K♠) 1/2652
⋮	⋮	⋮		⋮	⋮
(Q♠, A♡) 1/2652	(Q♠, 2♡) 1/2652	(Q♠, 3♡) 1/2652	··· ···		(Q♠, K♠) 1/2652
(K♠, A♡) 1/2652	(K♠, 2♡) 1/2652	(K♠, 3♡) 1/2652	··· ···	(K♠, Q♠) 1/2652	

Probability values for choosing 2 cards without replacement